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AY: 2025– 26	**Assignment No. 3**	Term- I

Department: Computer Engineering **Date:** 31.08.2025
Class: SE **Div:** E&F **Subject Name:** Operating Systems

Course Objectives

To understand and analyze process management concepts.

Course Outcomes

CO No.	Course Outcomes	BT Level
P203.2	Identify the mechanism for dealing with deadlocks and concurrency concerns.	**BT4**

Sr. No.	Question	CO	PI
Q1	Explain Mutex and Semaphore concept in detail.	**P203.2** BT4	1.6.1, 2.5.3, 3.5.2, 5.6.2, 9.4.1, 12.4.1
Q2	What is Dining Philosophers problem .	**P203.2** BT4	1.6.1, 2.5.3, 3.5.2, 5.6.2, 9.4.1, 12.4.1
Q3	What are necessary conditions for Deadlock in system?	**P203.2** BT4	1.6.1, 2.5.3, 3.5.2, 5.6.2, 9.4.1, 12.4.1
Q4	Explain Banker's Algorithm with example.	**P203.2** BT4	1.6.1, 2.5.3, 3.5.2, 5.6.2, 9.4.1, 12.4.1
Q5	**Example:** Considering a system with five processes P0 through P4 and three resources of type A, B, C. Resource type A has 10 instances, B has 5 instances and type C has 7 instances. Suppose at time t0 following snapshot of the system has been taken:	**P203.2** BT4	1.6.1, 2.5.3 3.5.2, 5.6.2 9.4.1, 12.4.1

Process	Allocation	Max	Available
	A B C	A B C	A B C
P_0	0 1 0	7 5 3	3 3 2
P_1	2 0 0	3 2 2	
P_2	3 0 2	9 0 2	
P_3	2 1 1	2 2 2	
P_4	0 0 2	4 3 3	

Q6	What are deadlock recovery techniques? Explain.	**P203.2** BT4	1.6.1, 2.5.3 3.5.2, 5.6.2 9.4.1, 12.4.1

Subject I/C